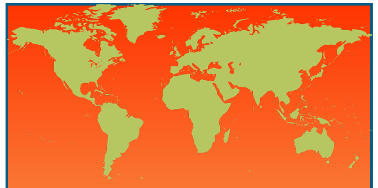
Digital Twin Practice

Demand and operations variability + AI reporting



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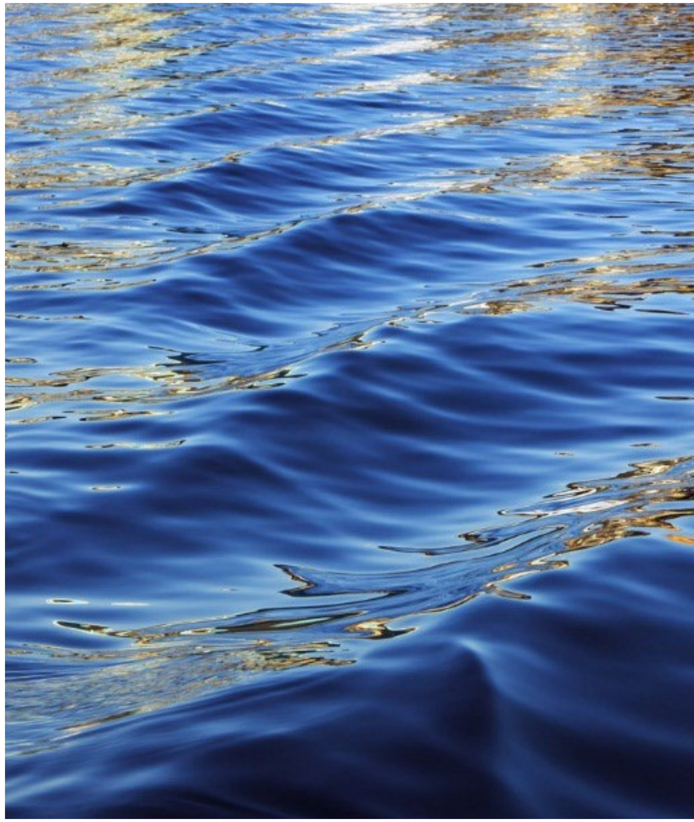
European University Alliance for Global Health



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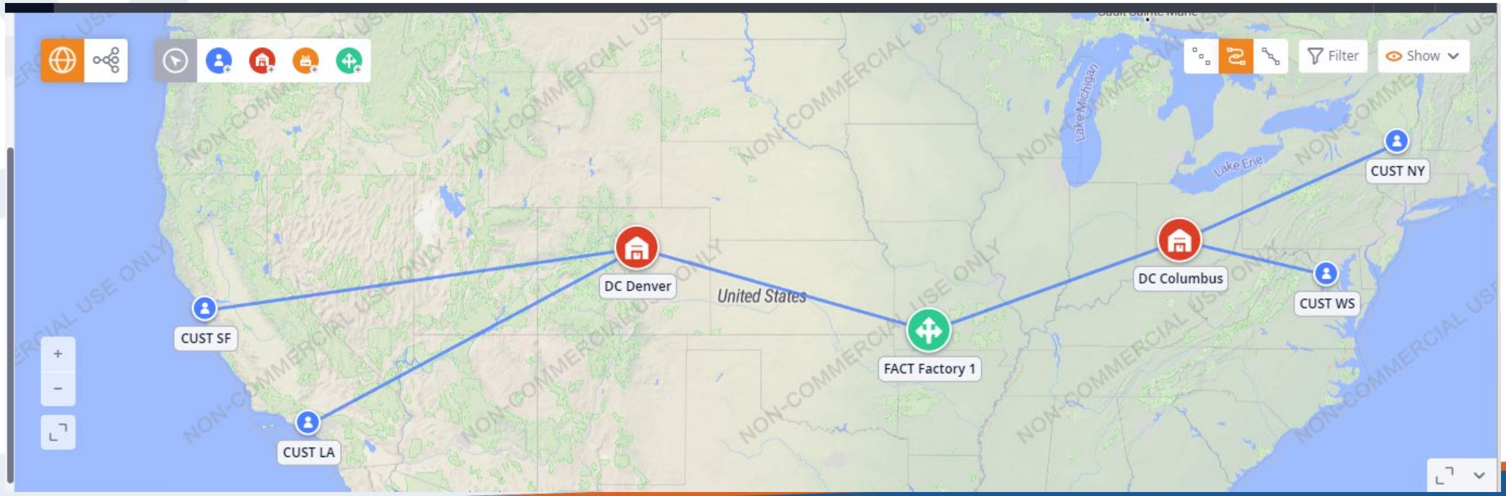
Learning objectives

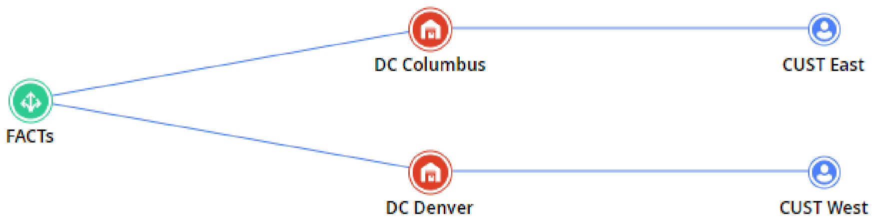
- Learn to simulate with variability:
  - Variable demands
  - Changing operations

- Quantitative Analysis
- Qualitative Analysis 🡐 Use an AI for this!

The Use Case

- ACME Inc. is producing plastic components from its factory in Springfield, Illinois, USA.
- It handles two distribution centers to serve USA market:
  - DC1 in Columbus, to serve East Coast Customers located in New York and Washington.
  - DC2 in Denver, to serve West Coast Customers located in San Francisco and Los Angeles.





The Use Case (II)

- The product is measured in pallets (uds):
  - Unit cost of 1,5K$ and a selling price of 2K$.
  - Pallet size is 1m3.
- Demand:
  - Periodic demand, with next day after interval, 1 day internal, amount of 200 uds.
- Inventory policies:
  - Min/Max with safety stock : s=5, S=10, ss=2
  - Initial stocks: 3.000 uds at factories and distribution centers.
  - Facility costs: 12 $/pallet/day.
- Transport policies:
  - By road, in trucks with a capacity of 20m3, and a mean speed of 50km/h.
  - Cost based on product*distance: 0,05cts/pallet/km.
  - Transport time: 5 days between factory and DCs , and 3 days between DCs and customers.
- Shipping policy is LTL, based on demand.

Prepare the dashboard for our analysis

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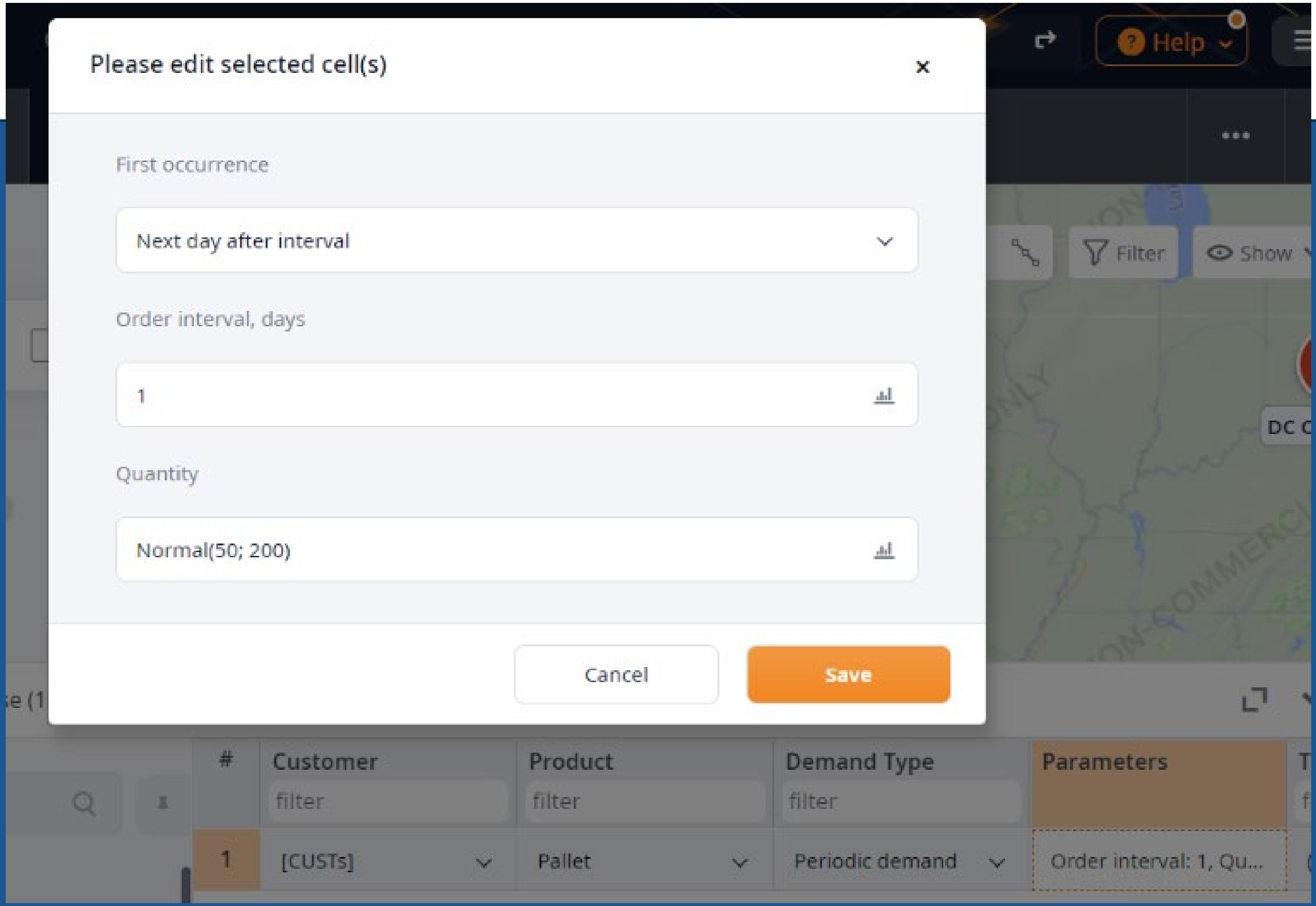
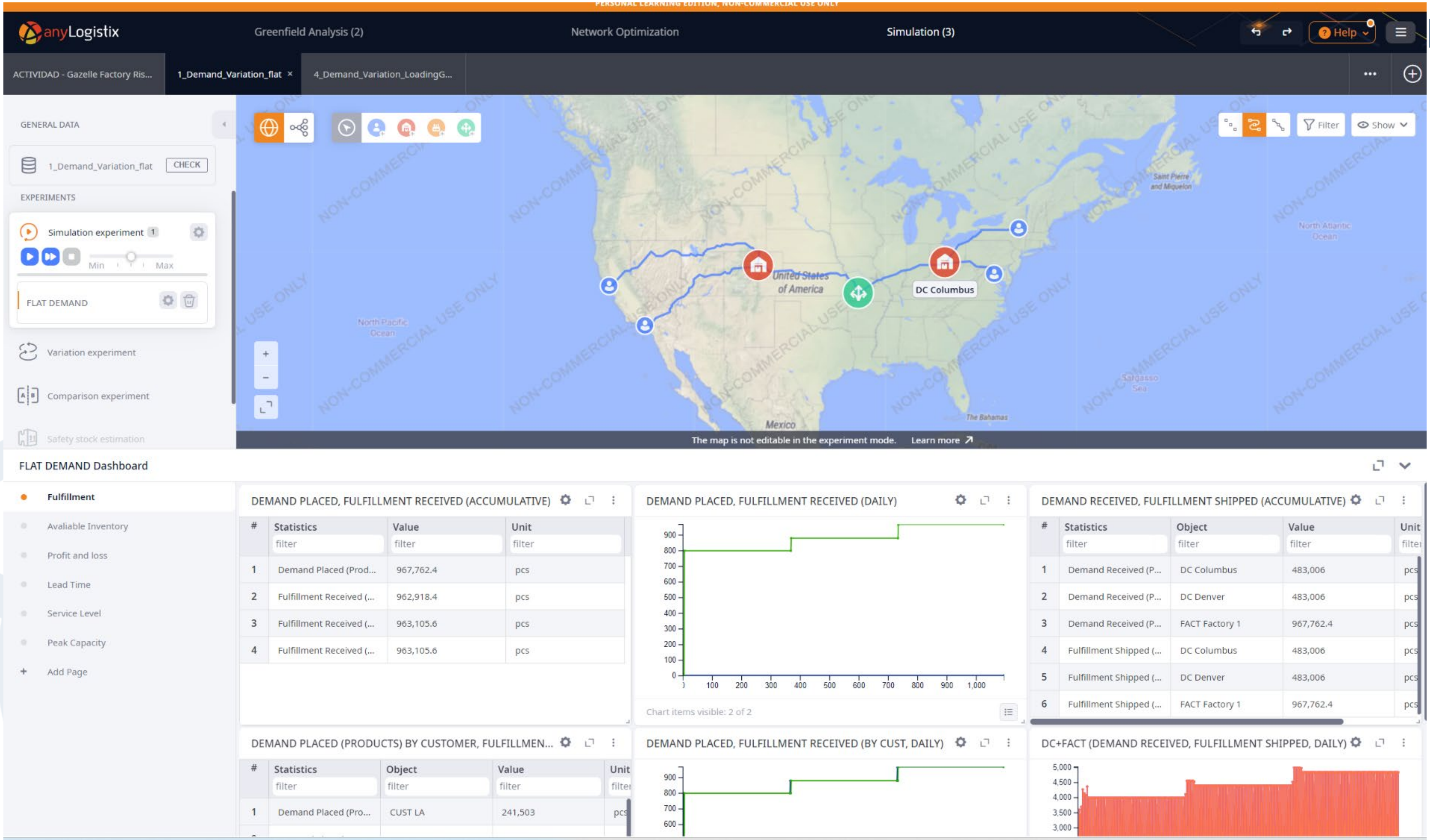
- Use “simulation experiments” (we are not designing).
- Dashboard to monitor activity:

Area	Purpose	KPIs
Fulfillment	How much demand are we receiving? Are we managing to meet it on time?	Demand Placed, Fullfillment received, Fullfillment received by customers, Fulfillment received on time.
Available Inventory	See how the inventory evolves in the factory and DCs.	Available Inventory, Daily Available inventory
Finantials	What is the profit generated from sales revenue and costs incurred? What is the breakdown by cost type and customer?	Revenue, Total Cost, Transportation Cost, Facility Costs
Lead Time	How much time takes to deliver the orders to the customers ?	Lead Time over time.
Service Level	Are we fulfilling our promise of on-time delivery to our customers?	ELT per product.
Peak Capacity	What is our peak capacity at our premises? Are we able to serve everything?	Peak Capacity per site

Scenario 1: the baseline



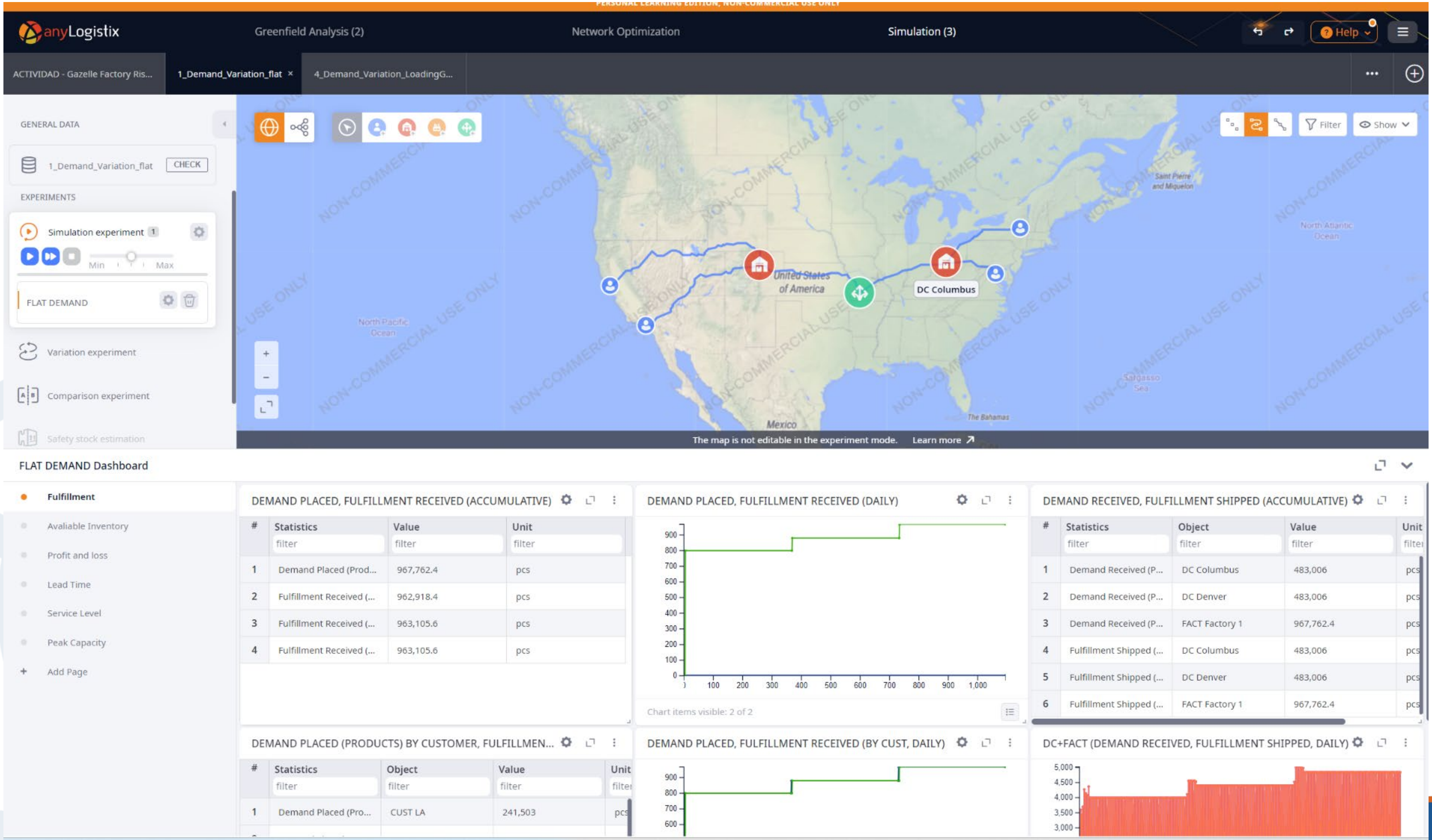


#	Statistics	Value	Unit
1	Demand Placed (Prod...	967,762.4	pcs
2	Fulfillment Received (...	962,918.4	pcs
3	Fulfillment Received (...	963,105.6	pcs
4	Fulfillment Received (...	963,105.6	pcs

#	Statistics	Object	Value	Unit
1	Demand Received (P...	DC Columbus	483,006	pcs
2	Demand Received (P...	DC Denver	483,006	pcs
3	Demand Received (P...	FACT Factory 1	967,762.4	pcs
4	Fulfillment Shipped (...	DC Columbus	483,006	pcs
5	Fulfillment Shipped (...	DC Denver	483,006	pcs
6	Fulfillment Shipped (...	FACT Factory 1	967,762.4	pcs

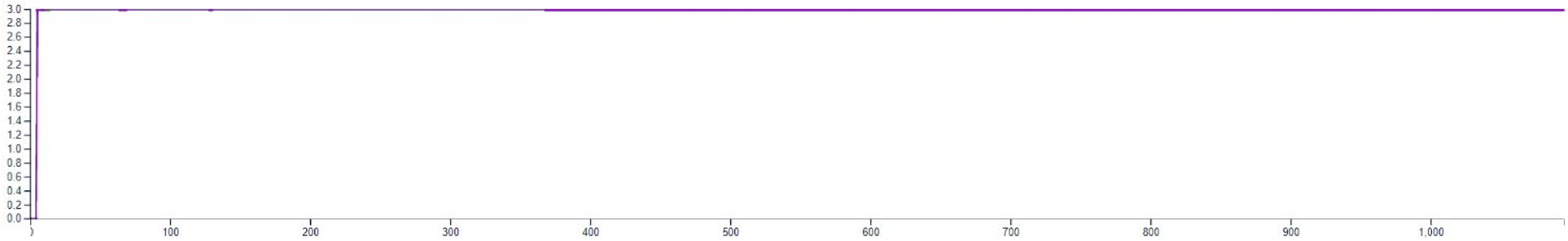
#	Statistics	Object	Value	Unit
1	Demand Placed (Pro...	CUST LA	241,503	pcs

Fulfillment

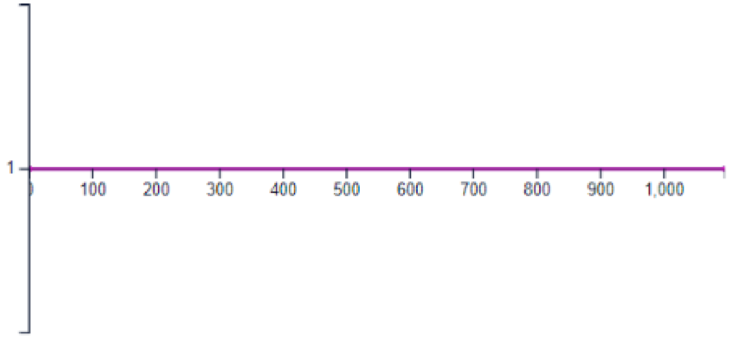


Service Levels

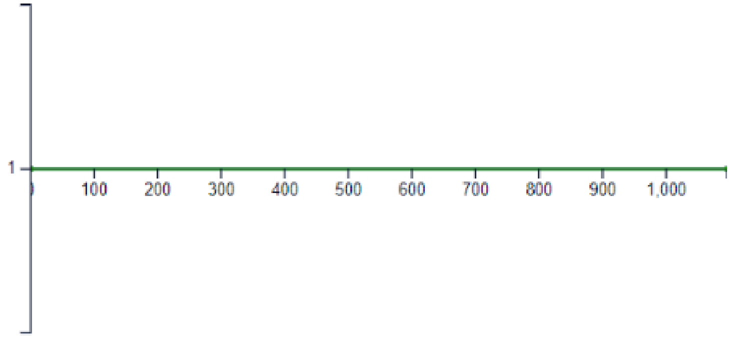
LEAD TIME



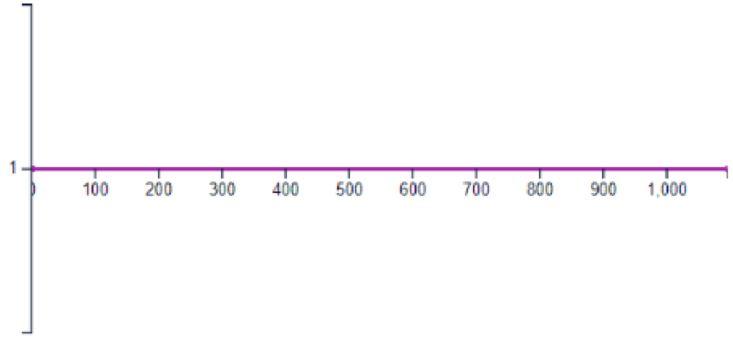
ELT SERVICE LEVEL BY PRODUCTS (PER PRODUCT)



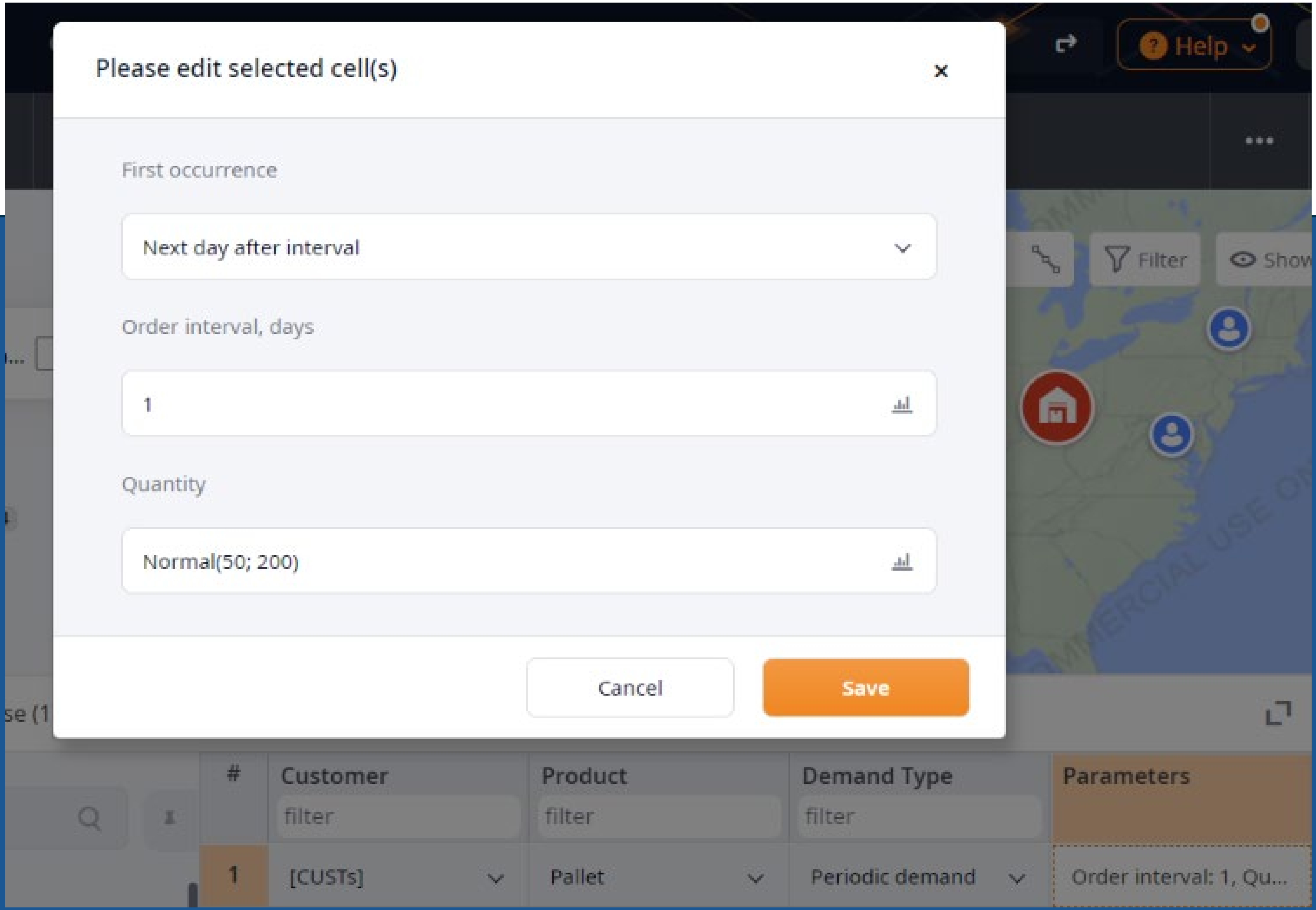
ELT SERVICE LEVEL BY PRODUCTS (PER SOURCE)

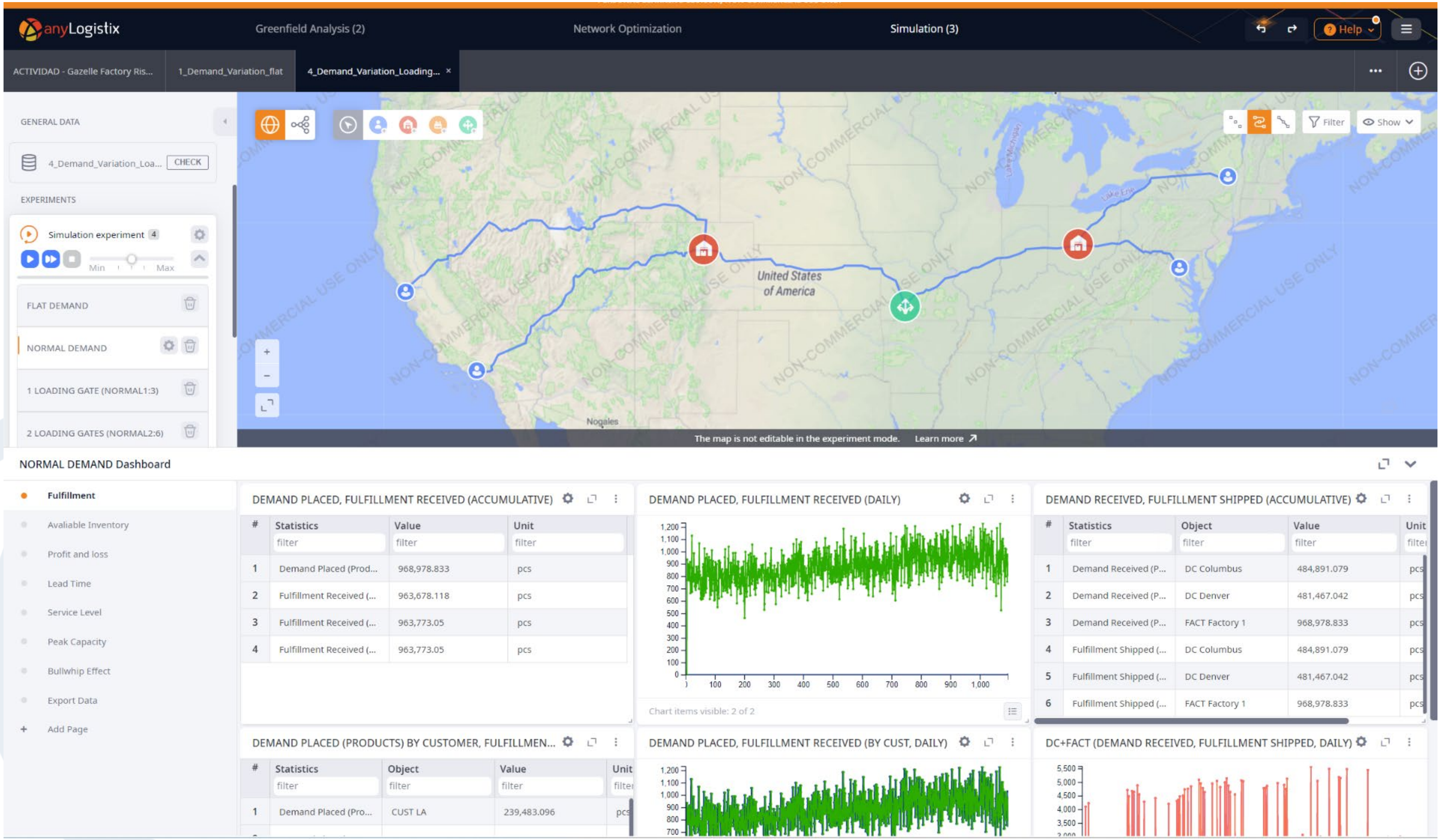


ELT SERVICE LEVEL BY REVENUE (PER OBJECT)



Scenario 2: variable demand





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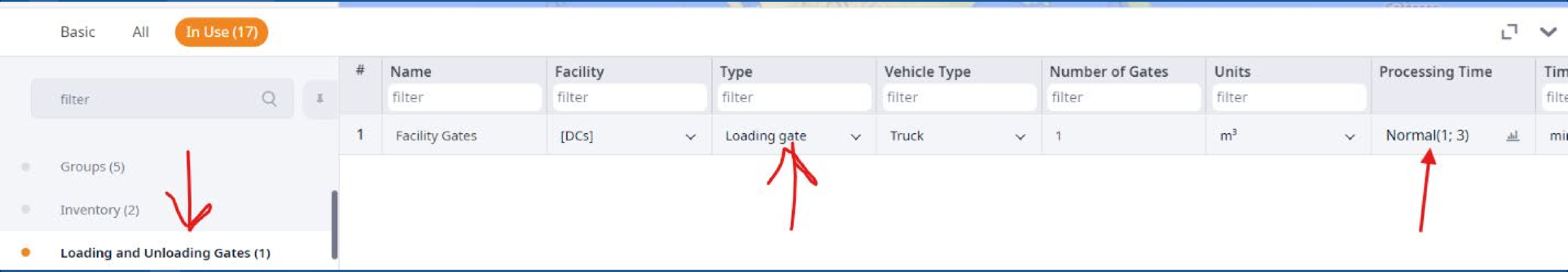
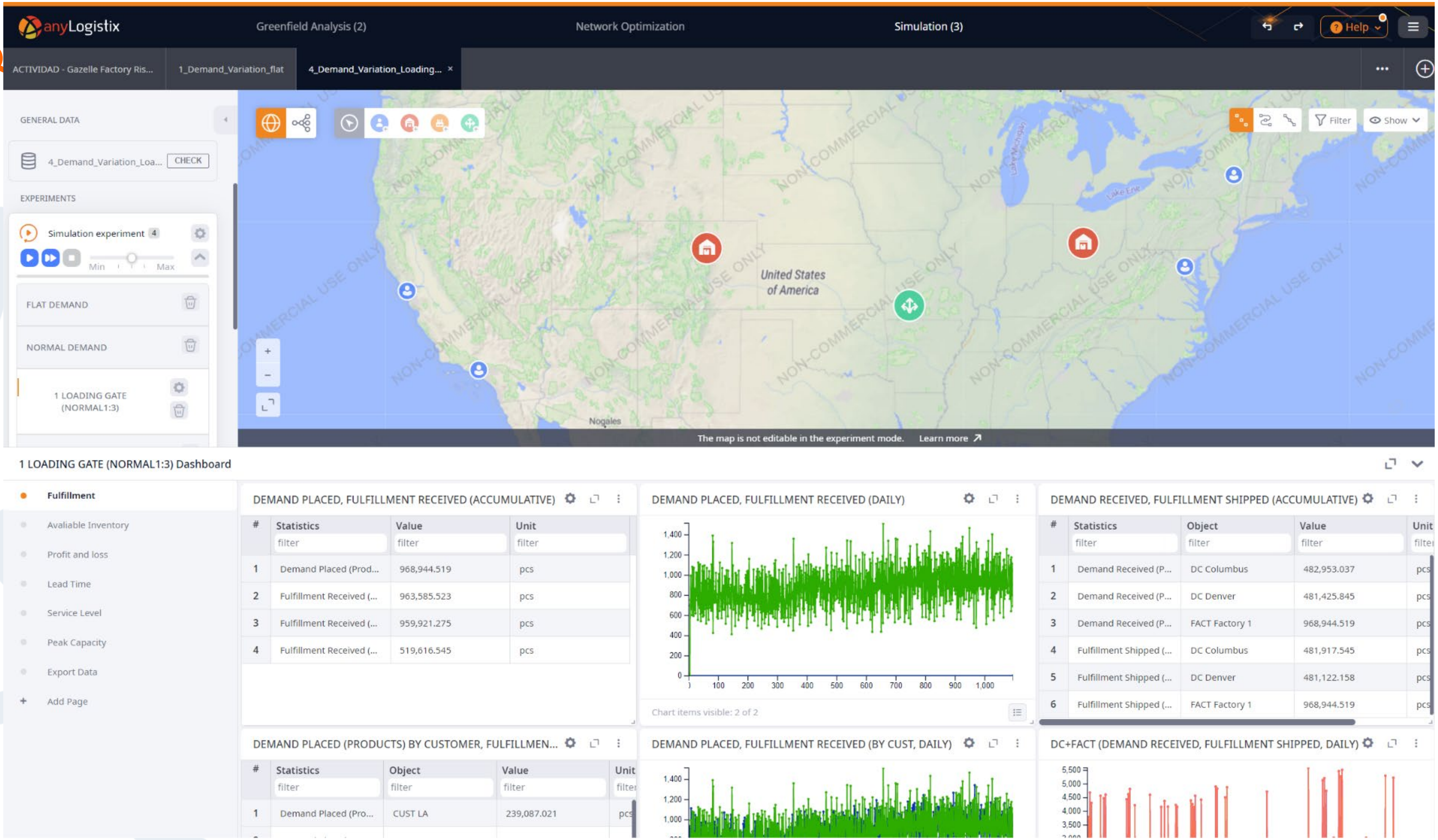
Scenario 3: variability in operations





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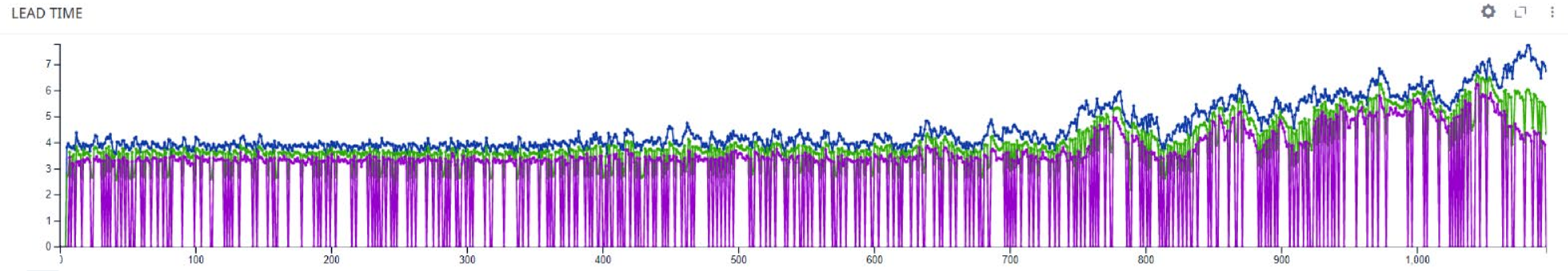
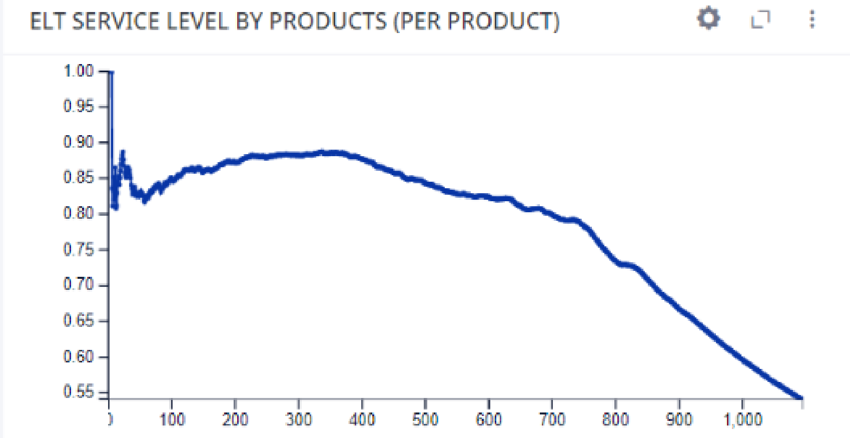
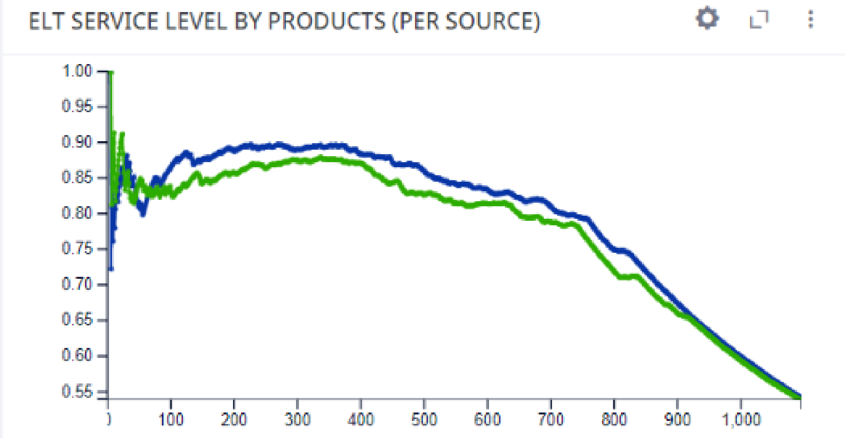
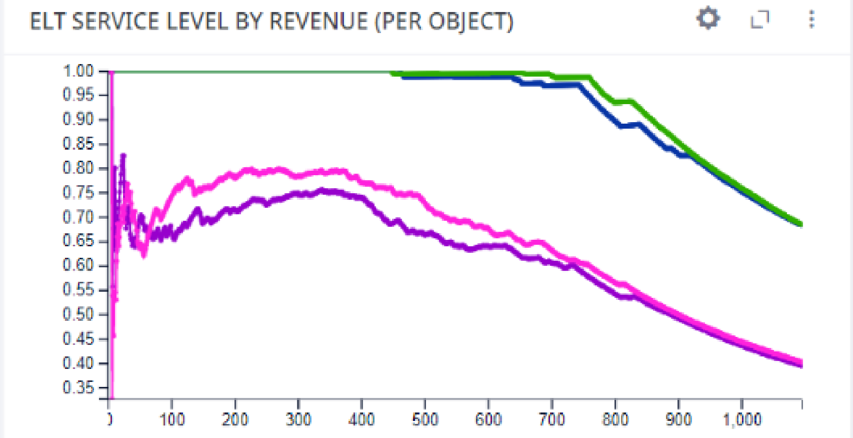
Service Level

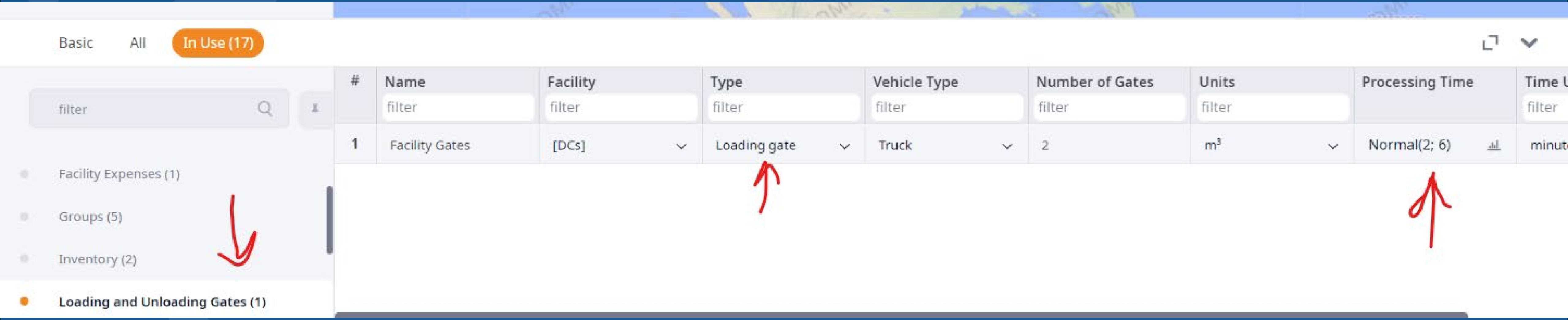




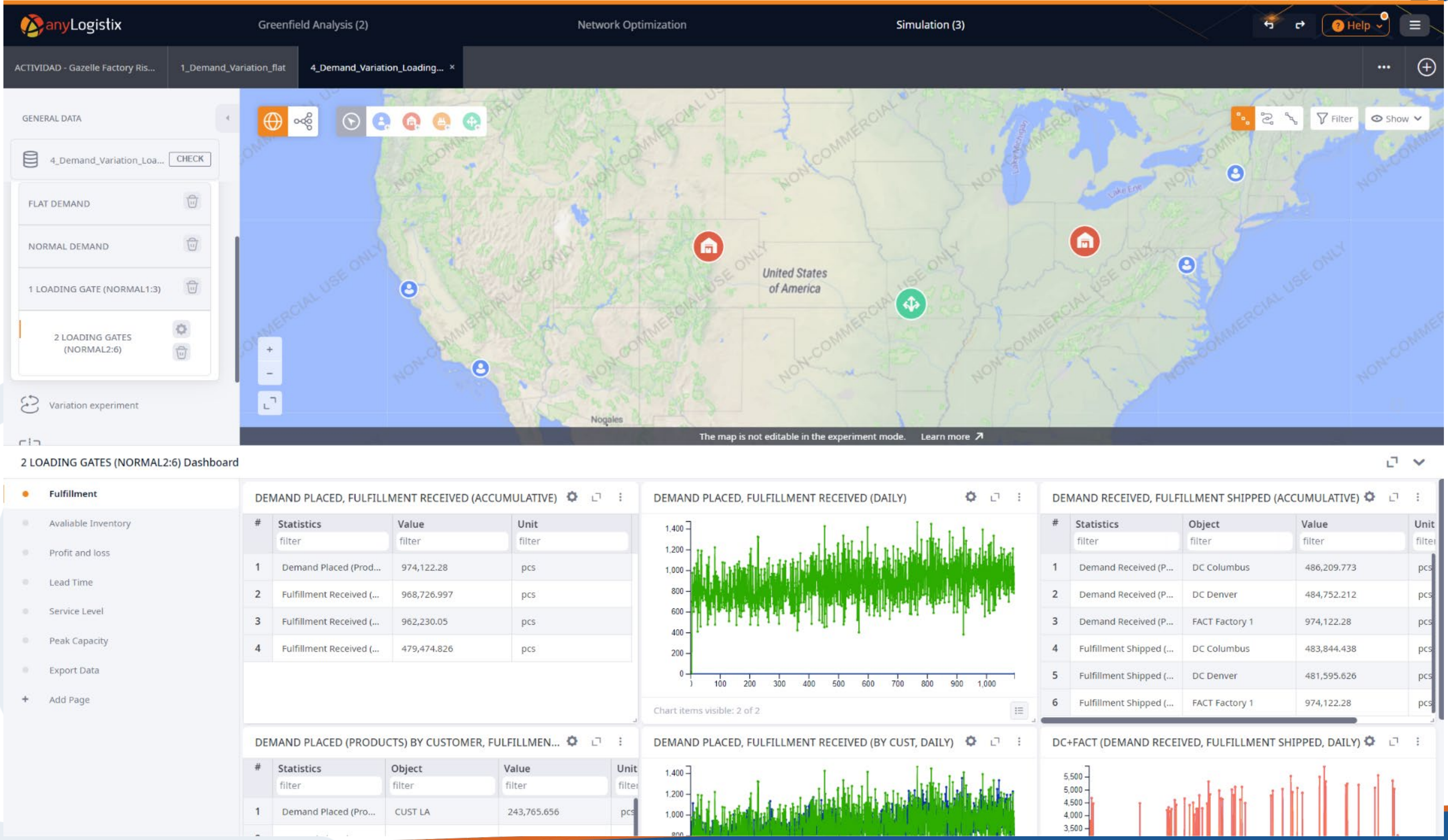




Scenario 4: increasing variability



Fulfillment



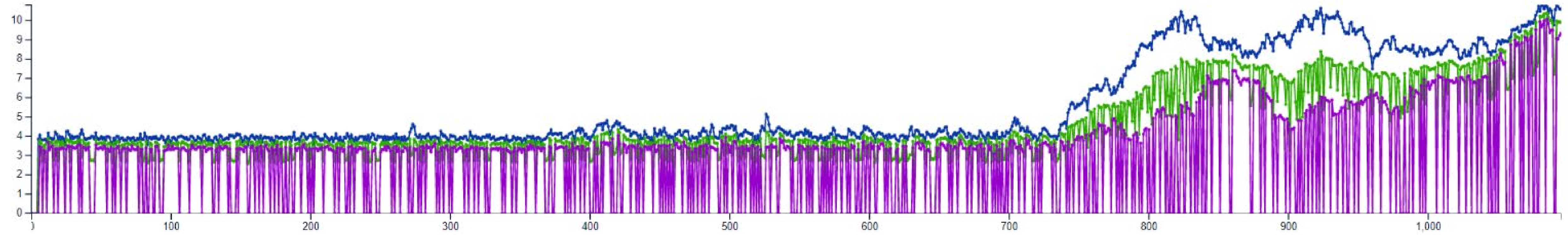
#	Statistics	Value	Unit
1	Demand Placed (Prod...	974,122.28	pcs
2	Fulfillment Received (...	968,726.997	pcs
3	Fulfillment Received (...	962,230.05	pcs
4	Fulfillment Received (...	479,474.826	pcs

#	Statistics	Object	Value	Unit
1	Demand Received (P...	DC Columbus	486,209.773	pcs
2	Demand Received (P...	DC Denver	484,752.212	pcs
3	Demand Received (P...	FACT Factory 1	974,122.28	pcs
4	Fulfillment Shipped (...	DC Columbus	483,844.438	pcs
5	Fulfillment Shipped (...	DC Denver	481,595.626	pcs
6	Fulfillment Shipped (...	FACT Factory 1	974,122.28	pcs

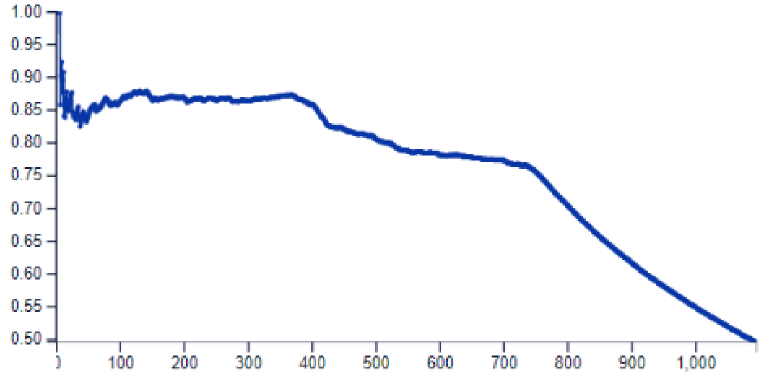
#	Statistics	Object	Value	Unit
1	Demand Placed (Pro...	CUST LA	243,765.656	pcs

Service Level

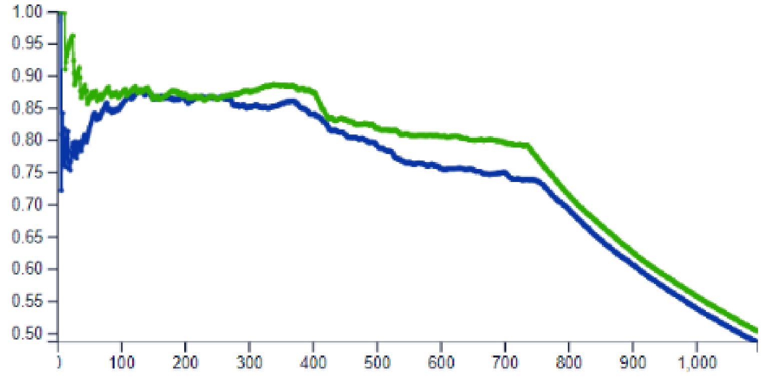
LEAD TIME



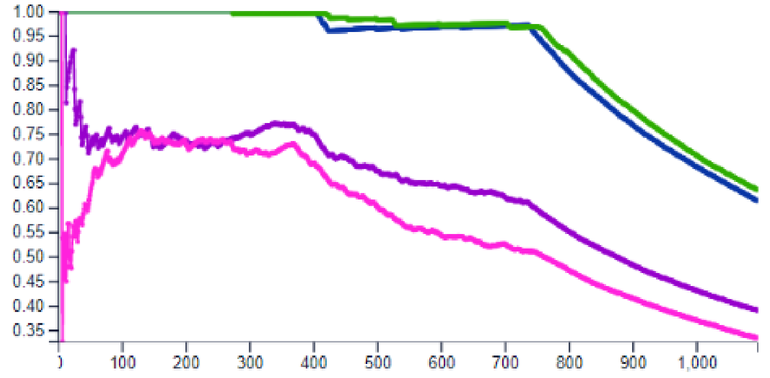
ELT SERVICE LEVEL BY PRODUCTS (PER PRODUCT)



ELT SERVICE LEVEL BY PRODUCTS (PER SOURCE)



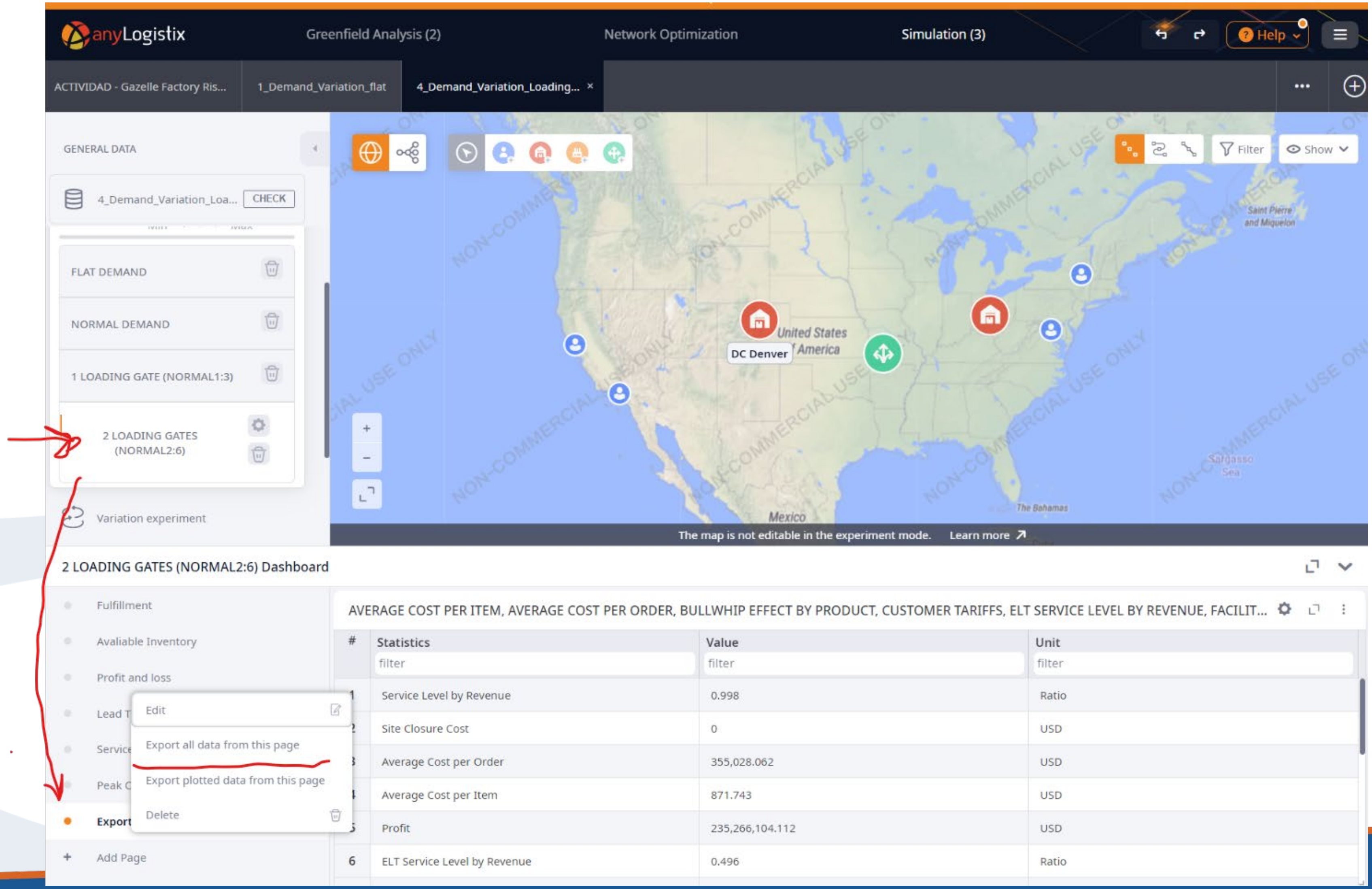
ELT SERVICE LEVEL BY REVENUE (PER OBJECT)



Use AI for Quantitative and Qualitative Analysis

How to compare all the scenarios?

- Export the experiment data and use an AI



The prompt

- In the role of "supply chain process analyst", I want to generate a quantitative comparative report (in English) for the result of 4 different simulations performed of a supply chain.
- Each of the simulations has dumped its results into an Excel sheet where the name of the simulation corresponds to the name of the Excel file without the extension ".xlsx".
- Each of the simulations collects its results within the "Table 1" sheet, in a data table from row 7 onwards, and columns A, B and C. Column A corresponds to the name of the indicator (KPI) obtained, column B corresponds to the value obtained for the KPI, and column C corresponds to the units of the KPI.
- For the analysis, you will perform the following steps:
  - 1. Create a comparison table where the rows of the column will correspond to the values obtained in column A, and as many columns as simulation scenarios are collected will be included, indicating in each column the VALUE of each KPI.
  - 2. Delete rows whose values are all "0".
  - 3. You will do a quantitative analysis for each simulation scenario including the assessment of all its KPIs, analyzing the pros and cons of each scenario. You will group the financial criteria first, then the service criteria and finally the rest of the criteria.
  - 4. Add a detailed root cause analysis to the report, justifying and explaining each of the variations. 5. You will upload the report as a professional-quality PDF file.
- the report title will be “Comparative Analysis of Supply Chain Simulation Scenarios” and include a summary page or executive overview at the beginning of the PDF

The report

- (This will be done as a deliverable activity for the students).